

The Optimal Use of Diagnostic Testing in Women with Acute Uncomplicated Cystitis

Stephen Bent, MD, and Sanjay Saint, MD, MPH

Acute uncomplicated cystitis is a common and costly disorder in women, and there is considerable variation in the diagnostic strategies currently used in clinical practice. Because the diagnosis of cystitis can be established in most patients using the history alone, the clinician's responsibility is to determine which patients require additional diagnostic testing. Patients with typical symptoms (i.e., dysuria, frequency, urgency, hematuria), without risk factors for complicated infection or pyelonephritis, and without a history of vaginal discharge, have a very high probability of cystitis and are appropriate candidates for empiric treatment. It is more difficult, however, to rule out infection in patients with suspected cystitis. Because the prevalence of culture-proven infection is very high in women who present with ≥ 1 symptom, and because the treatment threshold for this condition is low, a urine culture is generally required to rule out infection in patients with atypical symptoms, even in the presence of a negative dipstick test. In population-based, before-and-after studies, use of diagnostic algorithms has been shown to significantly decrease the use of urinalysis, urine culture, and office visits while increasing the percentage of patients who receive recommended antibiotics. These strategies have substantially reduced the cost of managing cystitis without an increase in adverse events or a decrease in patient satisfaction. Randomized controlled trials are needed to more closely examine the outcomes, costs of care, and patient satisfaction from different diagnostic and management strategies.

Acute uncomplicated cystitis is a common and costly disorder in women. Over 7 million annual office visits in the United States are attributed to this ailment,¹ which affects half of women at least once during their lifetime.² Yearly direct medical costs resulting from cystitis approximate \$1 billion in the United States.³ Despite its commonness, however, diagnostic testing for this disorder varies substantially.^{4,5} Whereas many authorities once recommended a thorough laboratory evaluation of women with dysuria that often included urinalysis, urine culture, and sensitivity testing before and after treatment, more recent studies suggest that these tests are rarely necessary in the setting of uncomplicated cystitis.^{1,6-8} Much of the variability in the clinical management of uncomplicated cystitis would likely be avoidable were evidence-based approaches to evaluation widely adopted.⁹ We thus provide an overview of the optimal use of diagnostic testing in women with symptoms of cystitis.

History

The history is critical in evaluating whether a woman has cystitis. Indeed, the diagnosis of cystitis can usually be made based on the history alone. The clinician's responsibility is carefully considering when additional diagnostic tests are needed. We describe the utility of different historical features below.

Although cystitis is a very common cause of dysuria (pain or difficulty when urinating), other disorders may cause this symptom. Specifically, there are 3 major infectious causes of acute dysuria in women: cystitis, urethritis, and vaginitis. Stamm and Hooton¹ provide a very helpful overview of the similarities and differences among these 3 disorders. In terms of pathogens, cystitis is usually caused by enteric gram-negative bacilli (e.g., *Escherichia coli*, *Proteus* species) or *Staphylococcus sapro-*

TABLE 1. Major Infectious Causes of Acute Dysuria in Women

Condition	Pathogen	Pyuria	Hematuria
Cystitis	<i>Escherichia coli</i> <i>Staphylococcus saprophyticus</i> <i>Proteus</i> spp. <i>Klebsiella</i> spp.	Usually	Sometimes
Urethritis	<i>Chlamydia trachomatis</i> <i>Neisseria gonorrhea</i> Herpes simplex virus	Usually	Rarely
Vaginitis	<i>Candida</i> spp. <i>Trichomonas vaginalis</i>	Rarely	Rarely

Adapted with permission from *N Engl J Med*.²

phyticus; urethritis is caused by *Chlamydia trachomatis*, *Neisseria gonorrhoeae*, or herpes simplex virus, whereas vaginitis is caused by candidal species or *Trichomonas vaginalis*. Whereas pyuria is usually present in both cystitis and urethritis, it is rarely present in vaginitis. Hematuria is occasionally present in cystitis but rarely present in urethritis or vaginitis. Urine culture is positive in cystitis but negative in urethritis or vaginitis. Finally, the symptoms of cystitis are usually acute, severe, and multiple; the symptoms of urethritis are usually gradual in onset and mild, and often include vaginal discharge or bleeding. Vaginitis usually leads to the symptoms of vaginal discharge or odor, dyspareunia, and pruritus (Table 1).¹

Several risk factors increase the probability that a woman has cystitis (Table 2).^{1,10-16} Similarly, the presence of the following symptoms, in addition to dysuria, increases the likelihood that cystitis is the diagnosis: frequency (frequent voiding of small volumes of urine), urgency (the feeling of having to urinate immediately), and hematuria (the presence of blood in the urine). On the other hand, women with vaginal discharge or irritation are much less likely to have cystitis and are more likely to have vaginitis.

Specifically, the utility of these symptoms has been evaluated in several studies.¹⁷⁻²⁰ In 8 urban clinics in New Zealand, Gallagher et al¹⁷ performed urine cultures on 130 consecutive women who presented with symptoms suggesting cystitis. They found that dysuria, fever, and hematuria were significantly more common in patients with documented bacterial infection compared with those with negative urine cultures. Using similar methods, Mond et al¹⁸ examined 83 women and found that dysuria, hematuria, fever, and suprapubic pain were more common in patients with cystitis, although they did not report the statistical significance of these findings. Among 821 women in the United States with symptoms suggesting either cystitis or vaginal infection, Komaroff et al¹⁹

TABLE 1. Continued

Urine Culture	Symptoms/Signs/Factors
$\geq 10^2$ – 10^5	Abrupt onset, severe symptoms, multiple symptoms (dysuria, increased frequency/urgency), suprapubic/low back pain, suprapubic tenderness on examination
$< 10^2$	Gradual onset, mild symptoms, vaginal discharge/bleeding, lower abdominal pain, new sexual partner, cervicitis/vulvovaginal herpetic lesions on examination
$< 10^2$	Vaginal discharge/odor, pruritus, dyspareunia, external dysuria, no increased frequency/urgency, vulvovaginitis on examination

TABLE 2. Important Risk Factors for Cystitis in Women

-
- Past history of cystitis, especially if recurrent
 - Recent sexual intercourse
 - Recent diaphragm and/or spermicide use
 - Unmarried
 - Lack of urination after sexual intercourse
 - Presence of asymptomatic bacteriuria
-

found that both dysuria and frequency were significantly more common in patients with positive urine cultures. More recently, a study of 54 patients found that the only symptom significantly more common in the subset of women with culture-proven cystitis was hematuria.²⁰ However, this study²⁰ likely lacked sufficient power to assess the diagnostic value of other symptoms, including dysuria, frequency, urgency, and abdominal pain.

Importantly, women presenting with the predominant symptom of dysuria could have urethritis or a vaginal disorder rather than cystitis. Thus, the elicitation of additional symptoms can be helpful in distinguishing between these disorders. Specifically, a history of vaginal discharge is a strong predictor of a vaginal disorder (i.e., vaginitis or urethritis), and the absence of this symptom greatly increases the probability of cystitis. Indeed, Komaroff et al¹⁹ found that only 9% of women with cystitis reported a history of vaginal discharge compared with 93% of women with vaginitis. Additionally, Wong et al²¹ reported that vaginal discharge was much more common in women with vaginitis (79%) than in women with a urinary tract infection (UTI; 26%).

Although individual symptoms can be useful, specific combinations of symptoms greatly increase the likelihood that cystitis is present. In a group of women with dysuria and frequency and without vaginal discharge, Komaroff et al¹⁹ found the probability of cystitis to be very high (77%). Conversely, in patients without dysuria and frequency but with vaginal discharge, the probability of cystitis was found to be very low (4%).¹⁹ These findings were largely confirmed in a study of 204 women in a university-based practice in which Berg et al⁸ found that 67% of patients with dysuria and frequency and without vaginal discharge had cystitis compared with only 14% of patients without dysuria and frequency but with vaginal discharge.

Thus, obtaining a history of the presence of certain symptoms (i.e., dysuria, frequency, hematuria) coupled with the absence of other symptoms (i.e., vaginal irritation or discharge) is a very useful strategy to diagnose cystitis. Often, a detailed history is all that is required.

Physical Examination

Compared with symptoms, specific features on the physical examination have been subjected to less rigorous evaluation. Although a few reports do provide data on the accuracy of certain physical examination signs for the presence of cystitis,^{21,22} in general, the physical examination is only marginally useful in the diagnosis of cystitis. In a retrospective chart review of 506 patients presenting to an emergency department with suspected UTI, Wigton et al²² found that the presence of costovertebral angle tenderness and the absence of vaginal discharge on pelvic examination were more common in patients with UTI. In a study comparing 53 women with symptoms of UTI to 139 women with a vaginal disorder, Wong et al²¹ found that vaginal discharge was more common in those with a vaginal disorder. In addition, although vaginal discharge was present in some women with cystitis, the discharge among those with vaginal disorders was substantially more copious. Thus, the available data indicate that the presence of vaginal discharge on physical examination has a strong association with vaginal infection and should alert the clinician to the possible presence of a sexually transmitted disease rather than a UTI. A pelvic examination is therefore obligatory in all women reporting vaginal discharge.

The physical examination may also be useful to distinguish between lower urinary tract (cystitis) and upper urinary tract (pyelonephritis) disease. Specifically, the presence of fever or costovertebral angle tenderness in a woman with dysuria likely indicates infection of the upper urinary tract rather than cystitis. Because it is difficult to define a “gold standard” for pyelonephritis, however, limited data exist that definitively assess the ability of physical examination findings to distinguish between cystitis and pyelonephritis. Given the pathophysiology of upper UTI, however, it seems reasonable to assume that fever and back pain are likely predictive of upper tract infection. Patients with these symptoms generally require urine culture and 7 to 14 days of antibiotics for presumed pyelonephritis. The development of a noninvasive gold standard test for pyelonephritis is an essential step to determining the value of various clinical findings for predicting the presence of upper UTI.

Urine Collection

The midstream clean-catch technique has long been considered the standard method by which to collect urine from women. Several investigators, however, have questioned whether the time and expense associated with this procedure are warranted.² This technique is somewhat

time intensive, because a woman must first clean her perineum with either a bactericidal wash or saline, spread her labia, discard the initial urine sample, and then collect a midstream sample in a sterile container. A recent randomized trial of 242 consecutive women (mean age, 21 years) presenting with symptoms of cystitis were randomized into 3 groups.²³ The first group (n = 77) urinated into a clean container without performing any cleansing; the specimen was not midstream. The second group (n = 84) used the proper midstream clean-catch technique. The final group (n = 81) used the midstream clean-catch technique and also used a vaginal tampon. Contamination rates for the 3 groups were very similar (29%, 32%, and 31%, respectively). Importantly, the authors considered the sample “contaminated” when the urine culture either revealed a mixed bacterial flora or when low levels (10^4 colony-forming units [CFU]/mL) of organisms commonly found on the skin and perineum (e.g., *Enterococcus*, *Staphylococcus epidermidis*) were recovered from culture. These authors conclude that the method of urine collection does not affect contamination rates and that collecting a specimen in a nonsterile container, without special instructions, will likely decrease resource utilization and patient embarrassment.²³ Confirmatory studies are required, however, before recommending that the midstream clean-catch technique be replaced by simpler methods of urine collection.

Although accurate diagnosis of cystitis may not necessarily rely on the use of a midstream clean-catch technique, it is likely related to appropriate transport of the urine specimen. Ideally, unrefrigerated urine should reach the microbiology laboratory within a few hours of specimen collection to avoid inaccurate results. Enteric bacteria may proliferate quickly at room temperature, thereby leading to false-positive results.² Refrigerated or preserved specimens, on the other hand, do not need to be analyzed within a few hours of specimen collection.

Urinalysis

Urinalysis can be used to confirm the presence of bacteriuria or pyuria and is usually the only laboratory test required to establish the diagnosis of acute uncomplicated cystitis in a symptomatic patient. Urinalysis and urine culture (including antimicrobial susceptibility tests) should always be obtained in patients with suspected pyelonephritis. Microscopic methods that may be used for determining the presence of bacteriuria include the Gram stain and microscopy of urinary sediment. The Gram stain, albeit currently rarely performed, is primarily used to detect bacterial presence in the urine, whereas microscopy of urinary sediment

TABLE 3. Primary Exclusion Criteria for the Group Health Urinary Tract Infection (UTI) Clinical Practice Guideline for Women

-
- Documented fever $>38^{\circ}\text{C}$
 - Symptoms of dysuria or urgency present for >7 days
 - Symptoms of vaginitis present (e.g., vaginal discharge or irritation)
 - Symptoms of abdominal pain, nausea, or vomiting
 - Gross hematuria in patients >50 yr
 - Presence of immunosuppression (e.g., current use of chemotherapeutic agents)
 - Diabetes mellitus
 - Known pregnancy
 - Chronic renal or urologic abnormalities, other than stress urinary incontinence (e.g., polycystic kidney disease, neurogenic bladder, renal failure)
 - Recent or persistent occurrence of urinary stones
 - Urinary catheterization or other urologic procedure within 2 wk
 - Discharge from hospital or nursing home within 2 wk
 - Treatment for UTI within 2 wk
 - Recurrent symptomatic UTI
-

is primarily used to detect pyuria. The most accurate method to detect pyuria is examining an unspun urine sample; the finding of ≥ 10 leukocytes per high-powered field is considered abnormal.²⁴

Importantly, the use of the leukocyte esterase test and the nitrite test (or urinary “dipstick” evaluation) has been found to be a useful indicator of cystitis. Often, this is the only evaluation needed for women with presumed cystitis. However, for patients who present with risk factors for complicated infection (Table 3), proper evaluation usually requires additional diagnostic tests, including urine culture and antimicrobial susceptibility testing.

Hurlbut and Littenberg²⁵ have identified and summarized 51 studies on the accuracy of dipstick urinalysis for the prediction of cystitis. These investigators provide summary receiver operating characteristic curves for various combinations of the nitrite and leukocyte esterase dipstick tests. Their analysis found that the nitrite-positive or leukocyte esterase-positive combination was the most accurate test. The point on the summary receiver- operating characteristic curve with the best accuracy represents a sensitivity of 75% and a specificity of 82%. Using other combinations of the nitrite and leukocyte esterase test will affect either sensitivity or specificity (i.e., requiring both to be positive decreases sensitivity and increases specificity). Others have reported even higher sensitivities (75% to 96%) and specificities (94% to 98%) for the leukocyte esterase test.²⁶ The nitrite test, however, is thought to lack adequate sensitivity when used alone.²⁶

Urine Culture

The gold standard for bacteriuria remains the quantitative urine culture. The amount of bacterial growth required before the urine culture is considered “positive” depends on whether a patient has symptoms of urinary tract infection. Among asymptomatic patients, bacteriuria is designated as significant when there are bacteria $\geq 10^5$ CFU/mL of voided urine in otherwise healthy patients. In women with clinical symptoms suggesting uncomplicated cystitis, a quantitative urine culture of $\geq 10^2$ CFU/mL is the most sensitive (95%) and specific (85%) test for diagnosing cystitis.⁷ In most women with presumed acute uncomplicated cystitis, however, urine culture is not required.

Treatment Threshold

The decision to treat a patient (for any condition) depends in part on the costs and benefits of treatment. Costs include both financial costs incurred by the patient and risks of adverse outcomes or side effects. Benefits include both symptomatic relief and the prevention of more serious diseases. The treatment threshold defines a probability of disease above which treatment is the preferred strategy (benefits of treatment outweigh costs of treatment) and below which no treatment is the preferred strategy (costs of treatment outweigh benefits of treatment).²⁷

From the perspective of a patient (which, importantly, does not include the societal costs of antibiotic resistance), the costs associated with a short course of antimicrobials for cystitis are generally low and the benefits (i.e., prevention of pyelonephritis, relief of symptoms) are generally high. This results in a treatment threshold that is rather low, because treatment thresholds are calculated by the algebraic expression of: cost/cost + benefit.²⁷ Although we are not aware of any formal calculation of a treatment threshold for the treatment of acute uncomplicated cystitis in women, a recent cost-utility analysis suggested that empiric treatment without urinalysis or urine culture is the most cost-effective strategy when the probability of cystitis in a woman with dysuria is greater than approximately 20% (Figure 1).²⁸

Using principles of evidence-based practice during a clinical encounter with a woman with suspected cystitis, the clinician first estimates the “pretest probability of disease,” or the probability that the woman has cystitis before any diagnostic tests are performed (including elucidating the presence or absence of specific findings from the history, physical examination, or results of dipstick evaluation, urinalysis, or urine culture).

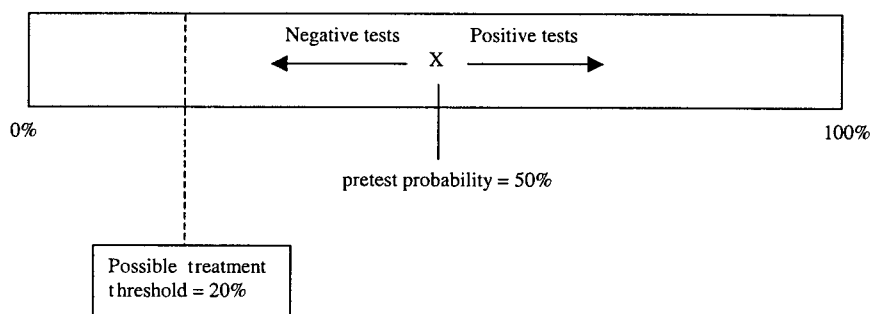


FIG 1. Treatment threshold and probability of urinary tract infection (see . text for explanation).

To determine the pretest probability of cystitis, we must evaluate the studies of women presenting with symptoms suggestive of cystitis (dysuria, frequency, hematuria) and find the prevalence of cystitis in those patients. In the largest 3 studies of cystitis examining only patients with symptoms suggesting cystitis, the prevalence of infection ranged from 45% to 59%.¹⁷⁻¹⁹ A rough approximation can be made that the pretest probability of cystitis in a woman who presents with some symptoms suggestive of urinary tract infection is about 50%.

The clinician then considers how the specific findings from the history, physical examination, and laboratory evaluation affect this probability of cystitis. As can be seen in Figure 1, positive tests increase the probability of cystitis and move the probability of disease closer to 100%. For example, if during the history the patient reports dysuria, frequency, hematuria, and no vaginal discharge, then each of these items increases the probability that the woman has cystitis. Conversely, if the patient reported vaginal discharge and irritation but denied having urinary frequency, these “tests” would move the probability of cystitis closer to 0%.

Unfortunately, the diagnostic accuracy of specific signs, symptoms, and laboratory tests for suspected cystitis has not been precisely defined in a quantitative manner. Therefore, estimating the “posttest” probability of disease must be done in a more general, qualitative fashion. Because patients who present with some symptoms of urinary tract infection have a relatively high probability of cystitis, then just a few other positive “tests” from the history (e.g., absence of vaginal complaints) indicate an even higher probability that cystitis is present. Such women can reasonably be treated for cystitis without any further diagnostic testing. If urinary dipstick evaluation is performed, the presence of a leukocyte

esterase or nitrite-positive test would further increase the probability of cystitis, supporting the strategy of treating without microscopic urinalysis or urine culture.

Unfortunately, when some items from the history and physical examination decrease the probability of cystitis (e.g., absence of frequency and hematuria), it is not clear whether this “posttest” probability is below the treatment threshold for cystitis. In effect, once a woman seeks care because of symptoms suggesting cystitis, it is rather difficult to establish that she does not have cystitis with any degree of certainty on the basis of history, physical examination, and dipstick testing. These patients should probably receive additional evaluation, including pelvic examination (if vaginal discharge is present), microscopic urinalysis, and/or urine culture.

Telephone-Based Management

Given the above discussion about probabilities of disease and importance of historical features, it is reasonable to consider telephone-based therapy for women with symptoms of acute uncomplicated cystitis. Recently, a small, randomized controlled trial compared telephone management with office evaluation in 72 women with suspected UTI.²⁹ Patients with symptoms suggestive of pyelonephritis, vaginitis, or cervicitis and patients with risk factors for complicated infections were excluded. All patients in the telephone management group and 33 of 36 patients in the office evaluation group were prescribed antibiotics to treat UTI. The study found that there was no difference in symptom scores or patient satisfaction with the 2 strategies, suggesting that telephone-based care is an appropriate strategy in carefully selected patients.

Similarly, a population-based, before-and-after study with concurrent control groups at 24 primary care clinics recently assessed the effect of a telephone-based cystitis guideline on resource utilization and on the occurrence of potential adverse outcomes (i.e., return office visit for cystitis or sexually transmitted disease, development of subsequent pyelonephritis).³⁰ After the development of a clinical practice guideline for acute uncomplicated cystitis in women aged 18 to 55 years at Group Health Cooperative of Puget Sound, a staff-model health maintenance organization in Washington,³¹ Saint et al³⁰ evaluated the clinical and economic consequences of the guideline. The cystitis guideline specifically advised health-care providers to manage low-risk women with symptoms of cystitis without a clinic visit or laboratory testing based on telephone evaluation by a primary care consulting nurse. The most important aspects of the guideline are described in Figure 2. Patients were excluded if they had any of the conditions listed in Table 2. A total of

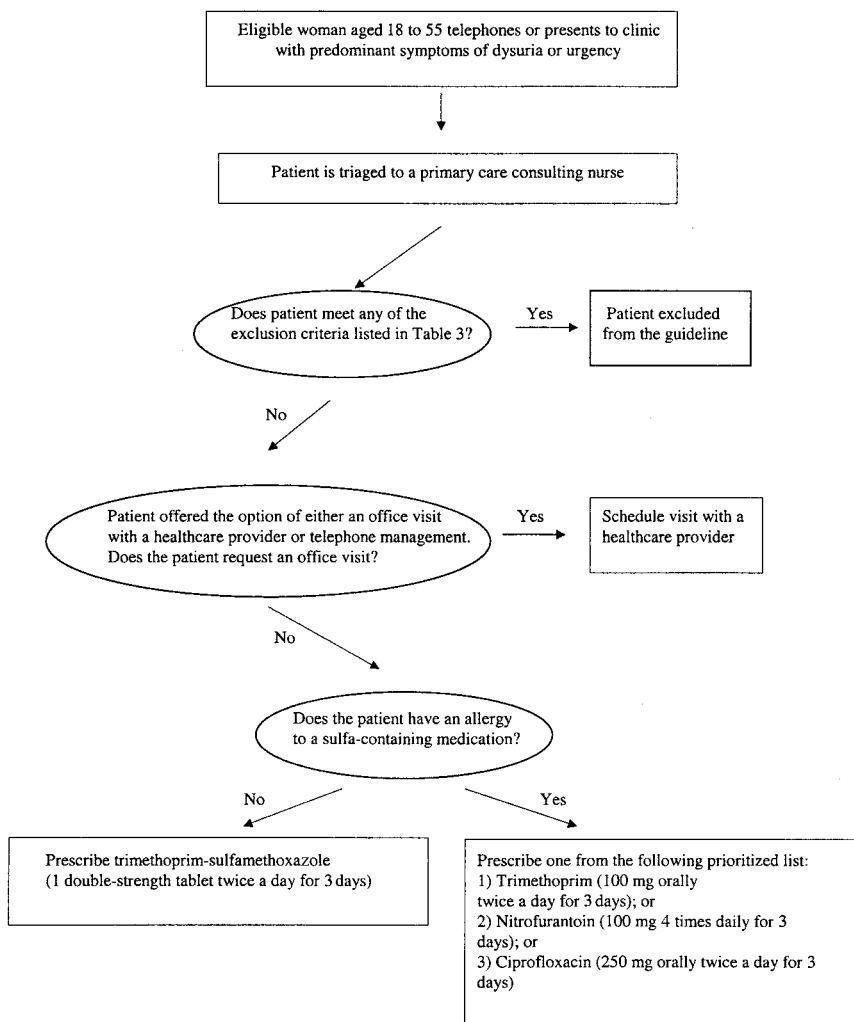


FIG 2. Group Health Cooperative algorithm for management of acute uncomplicated cystitis in women.

3,889 eligible patients were evaluated. As compared with baseline, guideline implementation significantly decreased the proportion of patients with presumed cystitis who received urinalysis (by 25%), urine culture (by 27%), and an initial office visit (by 33%), while significantly increasing the proportion receiving a guideline-recommended antimicrobial (3-fold). Adverse outcomes did not increase significantly, while cost savings of >\$300,000 were estimated. Finally, women who were man-

aged by means of the telephone-based guideline were extremely satisfied with the care they received.

The findings of Saint et al³⁰ are consistent with those of O'Connor et al,³² who also recently conducted an evaluation of 442 women managed by a guideline for presumed uncomplicated cystitis. Despite some differences in the guideline (e.g., no telephone-based decisions were made), the results also demonstrated a dramatic effect of the guideline on practice: an 89% decrease in the number of urine cultures and an 86% increase in recommended 3-day antimicrobial regimens prescribed after guideline implementation. With respect to adverse outcomes after guideline implementation, no significant differences were seen over the next 21 days in return visits related to cystitis, complications of cystitis (e.g., pyelonephritis), or competing diagnoses (e.g., sexually transmitted disease); 12.4% of preguideline patients had second visits versus 16.5% of postguideline patients ($P = 0.27$). The proportion of patients requiring a second course of antibiotics for cystitis also did not differ significantly (10.9% preguideline vs. 9.6% postguideline; $P = 0.74$).³²

Finally, the Cleveland Clinic has been using a telephone-based cystitis guideline since 1992 with reported success.³³ Although a detailed outcomes evaluation of this guideline has not been published, Campbell et al³³ do report a patient satisfaction study in 100 women in whom symptoms of presumed cystitis were managed by means of telephone-based evaluation and treatment. Specifically, 94% of women had symptom resolution, 100% reported being satisfied with the care received, and 98% stated that they would prefer to receive telephone-based management if symptoms of presumed cystitis recur rather than have an office visit with a health-care provider.³³ Unfortunately, of the 4 studies mentioned above,^{29,30,32,33} only Barry et al²⁹ assessed the proportion of patients with presumed cystitis who actually had positive urine cultures (64%). Thus, the positive predictive value of the cystitis guidelines remains uncertain, leading to concerns that these guidelines may result in overdiagnosis of cystitis and unnecessary antimicrobial therapy in some patients. Despite this important limitation, however, a telephone-based approach to the evaluation and management of women with presumed cystitis appears reasonable, at least in some clinical settings.

Diagnostic Algorithms

Several proposed algorithms exist for the approach to a patient with symptoms of cystitis.^{30,32,34} Most algorithms, like the one evaluated by Group Health Cooperative described in detail above,³⁰ initially attempt to include only those women with presumed acute uncomplicated cystitis.

Thus, patients with diabetes mellitus, recent urinary tract manipulation, possible pregnancy, receiving recent treatment for cystitis, or with other predictors for having a complicated infection are excluded from further consideration and require individualized management decisions. Once those with possible complicated infections are excluded, these algorithms focus on the clinical presence or absence of specific symptoms. Specifically, the presence of dysuria, urgency, and frequency coupled with the absence of vaginal irritation, vaginal discharge, or fever increase the likelihood of cystitis, such that further diagnostic testing may not be required. However, to further increase the likelihood of confirmation of cystitis, a dipstick urinalysis may be required. Once the presumed diagnosis of cystitis is made (with or without any laboratory evaluation), a 3-day course of an antimicrobial agent is usually prescribed (treatment for cystitis is discussed in another article in this supplement). Additionally, routine follow-up clinic visits or laboratory evaluation to confirm treatment success are no longer routinely recommended.

Barry et al²⁸ have formally evaluated the economic consequences of various diagnostic approaches to women with presumed cystitis. Using decision-analysis modeling, these investigators evaluated the cost-effectiveness of empiric treatment for all women with suspected cystitis compared with treatment based on dipstick testing, complete urinalysis, or urine culture. Using reasonable assumptions about costs and the frequency of side effects and complications, they found that empiric treatment was the most cost-effective strategy. As with most cost-effective analyses, this study did not include an assessment of the cost to society of increasing antibiotic resistance that might result from the use of an empiric treatment strategy. Also, whereas the empiric treatment strategy was the most cost-effective, other strategies (dipstick, urinalysis, and culture) were associated with greater utility (a measure of patient preference).²⁸ Nonetheless, this analysis suggests that empiric treatment is a cost-effective strategy for selected groups of patients.

Finally, a recent study by Gupta et al³⁵ evaluated the safety and feasibility of patient-initiated treatment in women with recurrent cystitis. The study recruited 172 women from a university-based practice with a history of ≥ 2 episodes of cystitis in the past year. The women were asked to self-initiate treatment with a 3-day course of fluoroquinolone antibiotics (specifically ofloxacin or levofloxacin) after obtaining a clean-catch urine sample for urinalysis and culture. During the 1-year study period, 88 of the women self-diagnosed 172 episodes of cystitis; 84% of these episodes showed a positive urine culture, and another 10% were thought to be probable cystitis based on the finding of pyuria on urinalysis.

Clinical cure was observed in 92% of the episodes of self-treatment, and there were no serious adverse events.³⁵ A similar study conducted by Schaeffer and Stuppy³⁶ examined 34 women with recurrent cystitis and found that 67 of 78 “self-diagnosed” infections (86%) were positive by urine culture; the cure rate was similarly high at 92%.³⁶ These findings suggest that highly educated women with recurrent cystitis are able to diagnose recurrent episodes reasonably well. Thus, self-treatment is likely safe and effective in certain populations.

Conclusions

Acute uncomplicated cystitis is an extremely common infection in women. Although diagnostic evaluation of this disorder has been shown to vary considerably, recent data provide important information to guide clinicians when evaluating women with presumed cystitis. It is critically important that health-care providers carefully consider when treatment can be initiated based on historical data alone and when additional diagnostic information is necessary. In women without risk factors for a complicated infection and in whom symptoms of pyelonephritis are not present, the symptoms of dysuria and frequency in the absence of vaginal complaints strongly indicates the presence of cystitis. Although dipstick evaluation can be performed to confirm the diagnosis of cystitis in such women, it may not be required. However, in women with atypical symptoms, or with evidence of pyelonephritis or vaginal discharge, a more thorough evaluation is necessary and may include pelvic examination, urinalysis, and urine culture. Although prospective studies of different diagnostic and management strategies are still needed, simplified cystitis guidelines and telephone-based treatment appear to be cost-effective and safe for carefully selected patients.

References

1. Stamm WE, Hooton TM. Management of urinary tract infections in adults. *N Engl J Med* 1993;329:1328-1334.
2. Kunin C. *Urinary Tract Infections: Detection, Prevention, and Management*. 5th ed. Baltimore: Williams & Wilkins; 1997.
3. Johnson JR, Stamm WE. Diagnosis and treatment of acute urinary tract infections. *Infect Dis Clin North Am* 1987;1:773-791.
4. Berg AO. Variations among family physicians' management strategies for lower urinary tract infection in women: a report from the Washington Family Physicians Collaborative Research Network. *J Am Board Fam Pract* 1991;4:327-330.
5. Wigton RS, Longenecker JC, Bryan TJ, Parenti C, Flach SD, Tape TG. Variation by specialty in the treatment of urinary tract infection in women. *J Gen Intern Med* 1999;14:491-494.

6. Stamm WE, Wagner KF, Amsel R, et al. Causes of the acute urethral syndrome in women. *N Engl J Med* 1980;303:409-415.
7. Stamm WE, Counts GW, Running KR, Fihn S, Turck M, Holmes KK. Diagnosis of coliform infection in acutely dysuric women. *N Engl J Med* 1982;307:463-468.
8. Berg AO, Heidrich FE, Fihn SD, et al. Establishing the cause of genitourinary symptoms in women in a family practice. Comparison of clinical examination and comprehensive microbiology. *JAMA* 1984;251:620-625.
9. Coleman RL. Promoting quality through managed care. *Am J Med Qual* 1992;7:100-105.
10. Fihn SD, Boyko EJ, Normand EH, et al. Association between use of spermicide-coated condoms and *Escherichia coli* urinary tract infection in young women. *Am J Epidemiol* 1996;144:512-520.
11. Fihn SD, Latham RH, Roberts P, Running K, Stamm WE. Association between diaphragm use and urinary tract infection. *JAMA* 1985;254:240-245.
12. Scholes D, Hooton TM, Roberts PL, Stapleton AE, Gupta K, Stamm WE. Risk factors for recurrent urinary tract infection in young women. *J Infect Dis* 2000;182:1177-1182.
13. Remis RS, Gurwith MJ, Gurwith D, Hargrett-Bean NT, Layde PM. Risk factors for urinary tract infection. *Am J Epidemiol* 1987;126:685-694.
14. Strom BL, Collins M, West SL, Kreisberg J, Weller S. Sexual activity, contraceptive use, and other risk factors for symptomatic and asymptomatic bacteriuria. A case-control study. *Ann Intern Med* 1987;107:816-823.
15. Foxman B, Geiger AM, Palin K, Gillespie B, Koopman JS. First-time urinary tract infection and sexual behavior. *Epidemiology* 1995;6:162-168.
16. Foxman B, Gillespie B, Koopman J, et al. Risk factors for second urinary tract infection among college women. *Am J Epidemiol* 2000;151:1194-1205.
17. Gallagher D, Montgomerie J, North J. Acute infections of the urinary tract and the urethral syndrome in general practice. *BMJ* 1965;1:622-626.
18. Mond N, Percival A, Williams J, Brumfitt W. Presentation, diagnosis, and treatment of urinary-tract infections in general practice. *Lancet* 1965;1:514-516.
19. Komaroff AL, Pass TM, McCue JD, Cohen AB, Hendricks TM, Friedland G. Management strategies for urinary and vaginal infections. *Arch Intern Med* 1978;138:1069-1073.
20. Nazareth I, King M. Decision making by general practitioners in diagnosis and management of lower urinary tract symptoms in women. *BMJ* 1993;306:1103-1106.
21. Wong ES, Fennell CL, Stamm WE. Urinary tract infection among women attending a clinic for sexually transmitted diseases. *Sex Transm Dis* 1984;11:18-23.
22. Wigton RS, Hoellerich VL, Ornato JP, Leu V, Mazzotta LA, Cheng IH. Use of clinical findings in the diagnosis of urinary tract infection in women. *Arch Intern Med* 1985;145:2222-2227.
23. Lifshitz E, Kramer L. Outpatient urine culture: does collection technique matter? *Arch Intern Med* 2000;160:2537-2540.
24. Stamm WE. Measurement of pyuria and its relation to bacteriuria. *Am J Med* 1983;75:53-58.
25. Hurlbut TA III, Littenberg B. The diagnostic accuracy of rapid dipstick tests to predict urinary tract infection. *Am J Clin Pathol* 1991;96:582-588.
26. Hooton TM, Stamm WE. Diagnosis and treatment of uncomplicated urinary tract infection. *Infect Dis Clin North Am* 1997;11:551-581.

27. Friedland D, Go A, Davoren J, et al. Evidence-Based Medicine: A Framework for Clinical Practice. 1st ed. Stamford: Appleton and Lange; 1998.
28. Barry H, Ebell M, Hickner J. Evaluation of suspected urinary tract infection in ambulatory women: a cost-utility analysis of office-based strategies. *J Fam Pract* 1997;44:49-60.
29. Barry HC, Hickner J, Ebell MH, Ettenhofer T. A randomized controlled trial of telephone management of suspected urinary tract infections in women. *J Fam Pract* 2001;50:589-594.
30. Saint S, Scholes D, Fihn SD, Farrell RG, Stamm WE. The effectiveness of a clinical practice guideline for the management of presumed uncomplicated urinary tract infection in women. *Am J Med* 1999;106:636-641.
31. Stuart ME, Macuiba J, Heidrich F, Farrell RG, Braddick M, Etchison S. Successful implementation of an evidence-based clinical practice guideline: acute dysuria/urgency in adult women. *HMO Pract* 1997;11:150-157.
32. O'Connor PJ, Solberg LI, Christianson J, Amundson G, Mosser G. Mechanism of action and impact of a cystitis clinical practice guideline on outcomes and costs of care in an HMO. *J Comm J Qual Improv* 1996;22:673-682.
33. Campbell J, Felver M, Kamarei S. "Telephone treatment" of uncomplicated acute cystitis. *Cleve Clin J Med* 1999;66:495-501.
34. Komaroff AL, Sawyer K, Flatley M, Browne C. Nurse practitioner management of common respiratory and genitourinary infections, using protocols. *Nurs Res* 1976; 25:84-89.
35. Gupta K, Hooton TM, Roberts PL, Stamm WE. Patient-initiated treatment of uncomplicated recurrent urinary tract infections in young women. *Ann Intern Med* 2001;135:9-16.
36. Schaeffer JA, Stuppy BA. Efficacy and safety of self-start therapy in women with recurrent urinary tract infections. *J Urol* 1999;161:207-211.